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Self-awareness:

A Key Component for Autonomous and Adaptive Systems

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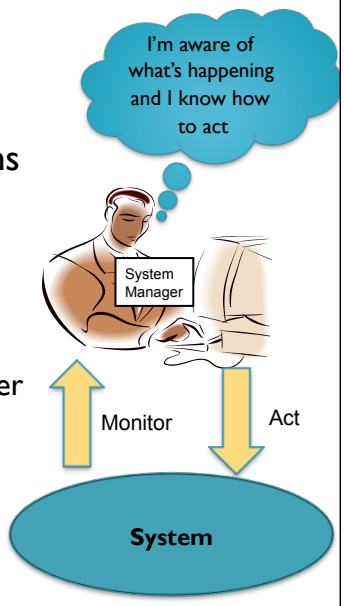
Outline

- Why?
  - Self-awareness as a key enabler to autonomous and adaptive behaviour
- What?
  - Concepts and implications of self-awareness
- How? Where? When? Who?
  - Challenges in modelling, engineering and building self-aware autonomic systems
- A Concrete Proposal:
  - the Knowledge Network Approach



Non Autonomous and Non-Adaptive Systems

- The system manager is human
- Increasing complexity of ICT systems
  - Mobility and ephemerality of nodes, variability of requests
  - Heterogeneity of comp/comm devices
  - Need of reliability and 24/7 availability
  - Need of "user-centric" adaptability other that system-level
- Challenging human-centered approaches to ICT systems management
  - Economically unbearable
  - Too much factors to be accounted for beings of limited rationality

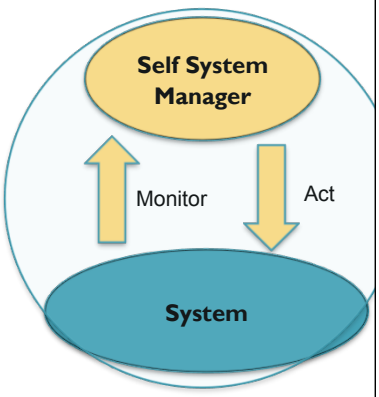


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Why Autonomous and Adaptive Systems?

- Autonomous → act in autonomy
- Adaptive → adapt to changes
- Autonomic = Autonomous + Adaptive
- Autonomic approaches consider that the role of human managers can be played by ICT systems themselves
  - Having them include the capability of self-monitoring and of self-acting
  - To realize a sort of control loop
- That's agents-based computing, in the end…

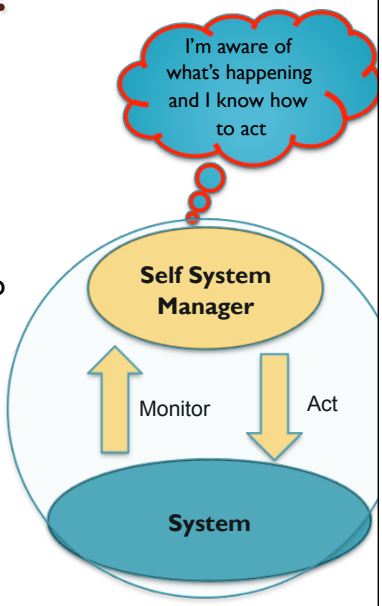


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Why Self-awareness?

- The capability of being autonomic requires:
  - The capability of understanding what's happening to themselves
  - The capability of what's happening to the environment
  - The capability of knowing what will happen in response to self-action
- That is:
  - "awareness" of the self and of the system in which the self-adaptive components
  - is a key ingredient of the autonomic recipe



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What has been done so far?

- Autonomic computing and communication researches have mostly focussed on architectural approaches
  - How to realize the autonomic control loop within ICT systems (e.g.: the MAPE-K model of IBM; the ACE-based component model of the CASCADAS project; self-* P2P networks architectures etc.)
  - What models, algorithms, tools, should be defined to promote decentralized and distributed autonomic systems (e.g.: nature-inspired algorithms, dynamic service composition, negotiation techniques, etc.)
- Agent-based computing
  - Reflects the above trends, although with a less-applied and too much theoretical endeavour (as it appear from the main AAMAS event)
  - Complex agent-based architectures, coordination models, etc.

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What about Self-awareness?

- Quite a neglected issue, despite its importance
- Most "user-centric" approach (pervasive computing, mobile and Web services)
  - Focus only on "context-awareness", as a way to adapt services/ applications to user context
- Most autonomic computing approach
  - Mostly reduce at assuming the capability of digesting information about the current system behaviour
- Still
  - The concept of "self-awareness" still requires to be properly unfolded in its various facets (abstraction levels, architectures, goals)
  - As it is the impact of "self-actions" on systems' behaviour and complexity (mostly understudied)

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What is "System"?

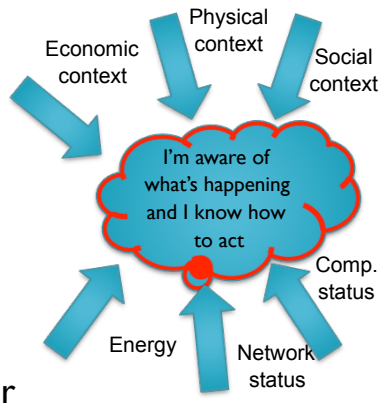
- What systems are we referring to? (e.g., where is self-awareness an issue?)
  - Large service/data centers
  - Internet and Telecom networks
  - Pervasive, mobile, and ad-hoc networks
  - Social networks and business ecosystems
  - System-level services vs. User-level services
- But the above distinction makes little sense
  - Systems are converging, being integrated, in need to coordinate with each other and work in an orchestrated way
- Need of general, foundational approaches, accounting for the current and future integration trends
  - Application-specific and technology-specific approaches could be of help, but would be definitely of limited impact

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Becoming aware of what?

- What are the situations of which a system should become aware of?
  - E.g. what should a system know to effectively self-express?
- Hard to confine
  - We can hardly abstract away from a large number of different situations that could impact the overall behaviour of the system and the satisfaction of the stakeholders
- Nothing like traditional layering or vertical integration can work
  - Yet dealing with all this information at a once can be very challenging



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How to Represent Data?

- Very diverse information sources and facts
  - Is there a way to uniformly represent it?
  - From user context to status of communication resources?
  - Can the need for a common representation be avoided?
- Can current Web-based languages by of some help?
  - XML, RDF, etc.
  - Or should be rather aim at simplest, lower-level, models
- How can the issue of semantics be addressed?
  - Should be aim for a common ontology?
  - Should be go for automated ontology integration and harmonization
  - Should we pragmatically go for emerging ontologies?

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But…are we talking only about data?

- Provided the availability of a sound data model, collecting data granules is not an issue (sensors' everywhere!)
  - FRANCO's iPHONE: GPS(lat, lon); Battery Level: 8%; Available Networks: WiFi, UMTS; Temperature: 3°; Activity: watching "Gone with the Wind"
- Getting a meaning out of diverse data is, indeed
  - Franco is close to home and will get there soon, as he is not expected to watch all the movie outside with such a freeze!
  - So I (the iPHONE) do not have to worry about saving battery power now by staying off-line, because Franco will soon have a chance to recharge me
- The key point is knowing what data to analyse and turning it into high-level knowledge about situations!



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How?

Data management and software engineering challenges

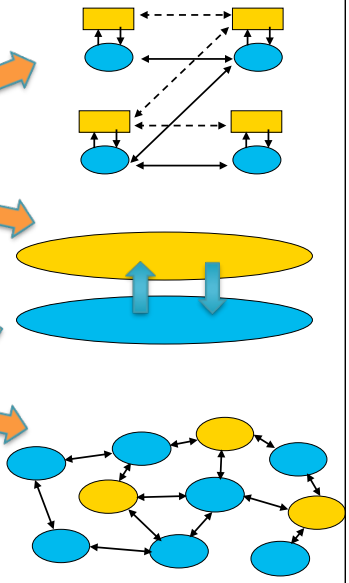
- Extracting knowledge from data
  - There is need of discriminating among possibly overwhelming amounts of data
  - There is need of correlating data granules with each other
  - There is need of recognizing situations of interest
  - ***Challenges***: common-sense analysis, historical analysis, knowledge overflow, etc.
- Dealing with multi-level abstractions
  - Not all components needs the same details
  - We should not forget the lessons of mainstream software engineering (abstraction! abstraction! abstraction!)
  - ***Challenges***: dynamic self-expression, to enable multiple degree of awareness within the same overall model and within the same software engineering abstractions



Where?

Architectural and algorithmic challenges

- Where should the capability of becoming self-aware reside?
  - Locally to each component
  - Globally to the system
- Where should information
  - Be collected, propagated, and how
  - Reside and be exploited
- Should all of this occur
  - In a separated plane?
  - As components' inherent behaviour?
- ***Challenges***: find the proper architectural and algorithmic trade-offs between
  - Local vs global
  - Separation vs integration of concerns



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When?

Prediction and evolution challenges

- Short-term
  - Knowing what is happening is important and can be effective to correct situations
  - Knowing what will happen is important to prevent incorrect situations to occur
  - ***Challenge***: Self-aware systems should become "future-aware"
- Long-term
  - How is my technological context (kinds of computing devices, type of connectivity, services provided) changing?
  - How is my context of usage (social/economical) changing?
  - ***Challenge***: defines models and architectures so as to be "future-proof", ensure long-lasting effectiveness and evolvability

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Who?

Business models and security challenges

- Who can act on the system?
  - Answer: everybody!
  - Open production models (prosumption)
  - Long tail of needs and requirements (accommodate diversity)
  - ***Challenge***: how can I protect my system and its data without erecting walls yet accommodating diversity and openness to prosumers? How can the system become aware of privacy/ security issues?
- Who controls the controllers?
  - If all autonomic behaviour is based on self-awareness
  - How can the system become aware of self-unawaress?
  - ***Challenge***: ensure autonomic behaviour in the self-awareness-related parts of the system, yet avoiding infinite levels of recursion (meta-meta-meta-levels)

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State of the Art

- The various issues related to self-awareness are indeed attracting increasing attention in different areas
- User-centered services:
  - moving from simple context-recognition to activity-based recognition
- Autonomic computing and communication services:
  - Moving from context-aware systems to situation-aware systems

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Our own contribution: The "Knowledge Networks" Approach

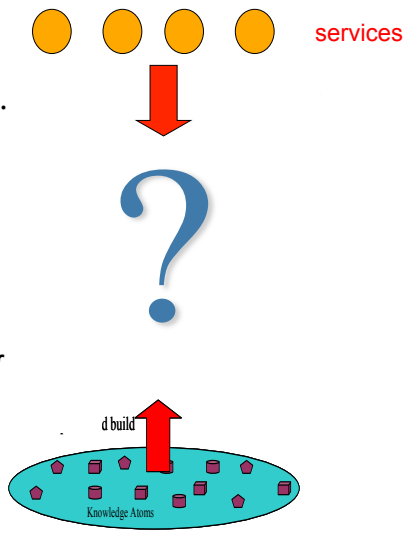
- A key result of the CASCADAS project (FP6 FET initiative on situated and autonomic communications)
- Based on ACEs, sort of autonomic agent model defined within the project for the development of general-purpose pervasive autonomic services defined
- Identify models, algorithms and tools for
  - Organize contextual knowledge and produce new knowledge from available one
  - Make it available to ACE-based services so as to enable them to become "situation-aware" in an "easy" and effective way
  - Enabling perceiving situations in a service-dependent way
- Implementing KN services with ACEs in their turn

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The Challenge of CASCADAS "Knowledge Networks" (KN)

- Sensors of any type anywhere
  - Sensor networks, tags, profilers, service descriptions, Web 2.0, etc.
  - Generating overwhelming amounts atomic units of knowledge (Knowledge Atoms)
- How can services use them to become "situation-aware"?
  - Hard to manage huge amounts of heterogeneous knowledge atoms (data, more than knowledge)
  - They represents individual contextual facts, not situations
  - Data not application-specific



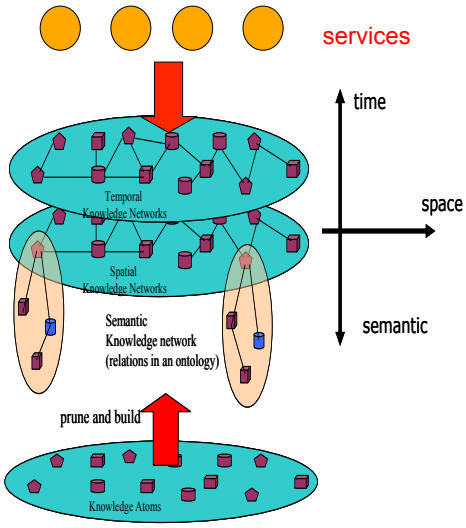
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The KN Reference Architecture

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- We have to
  - Make data homogeneous
  - Prune (aggregate/compress) data to make it manageable by services
  - Identify relations between atoms to identify "situations" → networking knowledge
  - Generate "new knowledge"
  - All of which in a "self-*" way
- There is not a single KN
  - Knowledge aggregated along spatial, temporal and semantics dimensions
  - Application-specific needs
  - Multiple KN views

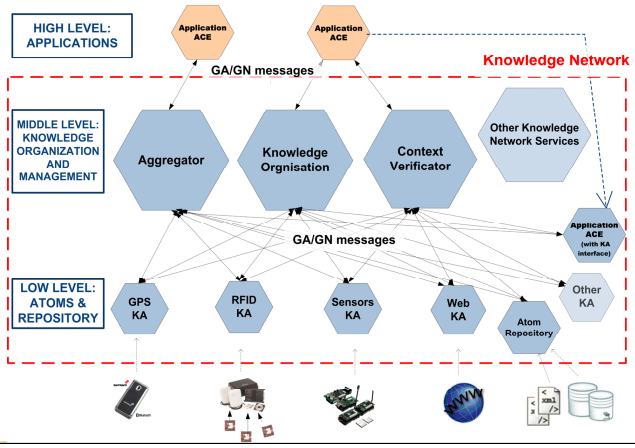




The KN Software

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- Implemented as a set of ACE libraries interacting wi the GA/GN protocol
  - Extensible (even at run-time)
  - Distributable





Towards the W4 Data Model

- How can we provide a uniform representation of KA data in the KN? (i.e., facts about the world, that is knowledge about it)
  - Coming from diverse devices (sensors, tags, cameras, and why not Web 2.0 fragments such as geo-tags)
  - Expressing different levels of observations (e.g., a single light sensor vs. a camera)
  - Expressing in a uniform way different redundant perspectives on the same fact (e.g., a WiFi vs. a GPS localization information)
  - Easy to be accesses and manipulated, aggregated (by both services and by within the world model)
- Key guidelines
  - Keep it simple, intuitive, and computable
- How do we usually characterize facts about the world?
  - A possible idea: the W4 model

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The W4 Data Model (1)

- Someone or something (*Who*) does some activity (*What*) in a certain place (*Where*) at a specific time (*When*)

- **Who** is the subject. It is represented by a string with an associated namespace that defines the "kind" of entity that is represented.
  - "person:Gabriella", "tag:tag#567", "sensor-region:21"
- **What** is the activity performed. It is represented as a string containing a predicate-complement statement.
  - "read:book", "work:pervasive computing group", "read:temperature=23".
- **Where** is the location to which the context relates.
  - (longitude, latitude),
  - "campus", "here"
- **When** is the time duration to which the context relates
  - 2006/07/19:09.00am - 2006/07/19:10.00am
  - "now", "today", "yesterday", "before"

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The W4 Data Model (2)

- All knowledge atoms represented in this way

- Gabriella is walking in the campus' park, and her GPS-equipped PDA generates this atom:
  - **Who**: user:Gabriella
  - **What**: speed:4Km/h
  - **Where**: @45.910108, 13.50721
  - **When**: 14.21.56 March 5

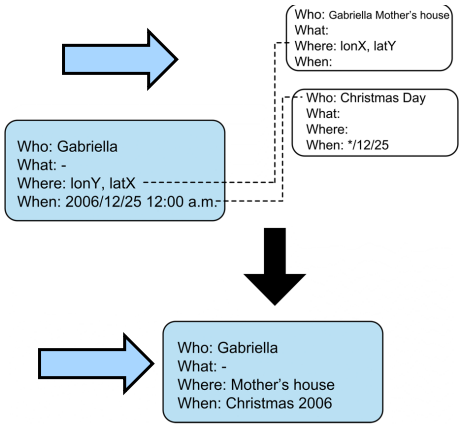
- Query: Is there fire around?
  - **Who:** sensor:*
  - **What:** temperature > 60
  - **Where**: circle,center(lonY,latX),rad:500m
  - **When:** now

- Key features
  - Tolerates partial knowledge (not fully filled atoms)
  - Natural support for context-aware queries ("here", "now")
  - Possibility of generating atoms merging different sources
  - Data source independence

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Knowledge Networking and Aggregation

- W4 knowledge atoms can be related to each other
  - To represent relations between pieces of data and enable navigating related facts of the world
  - To support a better navigation of services in knowledge
  - And more "cognitive" forms of self-organization in services
- Starting from the networking
  - Related knowledge atoms can be aggregated according to application-specific mechanisms
  - To create higher-level W4 knowledge atoms from low-level W4 atoms



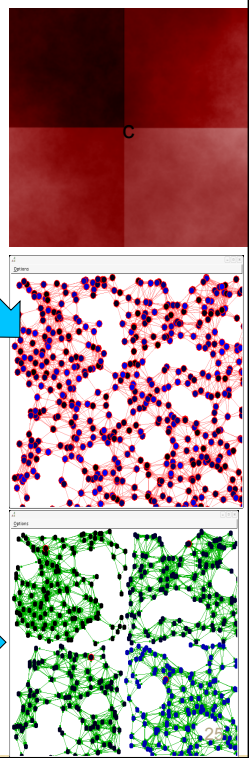


Knowledge Networking and Aggregation Examples:

Virtual Macro Sensors

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- Assume a sensor network embedded in an environment characterized by varying patterns of some phenomena
  - Physical connection between sensors based on wireless connectivity
- Gradually, close sensors recognize local, knowledge-based, logical relationships
  - A distributed KN algorithm is running in the network as a sort of *"background noise"*
  - With the goal of partitioning the sensor network into regions characterized by similar patterns (as an overlay of KN links)
  - Abstracting from the specific density/structure of the sensor network
- Then, a logical partitioning emerges based on the environmental patterns
  - Knowledge network of sensors nodes

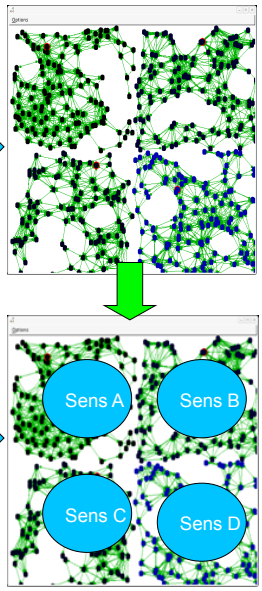




Eventually…

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- Once Regions are Formed
- Aggregation of data can take place on a per region basis
  - Averaging data over the region
  - Computing regional functions
  - All sensors in the region acting as a single entity
- The result is a network of "virtual" macro sensors
  - Services can start perceiving the sensor network as composed of a finite number of macro sensors
  - Abstracting from the actual structure and density (potentially continuum) of the physical network
  - Rather focussing on the real environmental characteristics



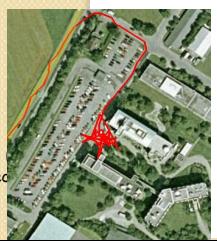


Knowledge Networking and Aggregation Examples:

The Whereabouts Diary

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- The whereabouts diary is a useful application-specific knowledge networks tool (i.e., to support location-based services), based on:
  - Automatic spatio-temporal-semantic networking of user data with the W4 model
  - W4 Data aggregation to generate higher level knowledge
- Starting from a simple set of GPS reading
  - Mining the Web for possibly related W4 atoms (inverse geo-coding, white and yellow pages)
  - Getting advantage of environmental sensor networks around
  - Mining the past history of W4
  - Applying probabilistic analysis…



Who	What	Where	When
Franco	Speed: 1Km/h	@11°16'43.17", 48° 5'11.75"N	March 5, 2008, 8:17.23am
Franco	Speed: 3Km/h	@11°16'43.21", 48° 5'11.77"N	Sept. 20, 2007, 8:17.33am
…	…	…	…



Eventually…

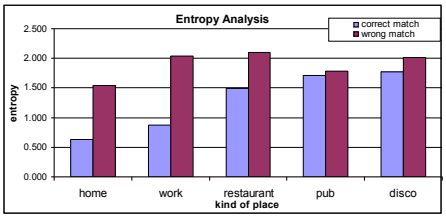
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Who	What	Where	When
Franco	Sleeping	Hotel Diplomatic	Sept 8, 2008, 00:27am – 8.25am
Franco	Traveling	Turin	Sept 9, 2008, 8.26am– 8.42am
Franco	Work	MALLOW Workshop	Sept 9, 2008, 8:43am-16:47pm
…	…	…	…

We can automatically generate a high-level "knowledge-based" diary of users' activities

With high accuracy

Correct	Incorrect		
	Wrong	False negative	False positive
80.6%	0%	17.3%	2.1%





Conclusions

- Self-awareness is a key ingredient to promote autonomic behaviour in ICT systems
- Yet, there are many foundational aspects and challenges to be addressed to effectively achieve it
  - Extracting knowledge from data
  - Dynamic self-expression of proper abstraction levels
  - Global vs. local awareness
  - Prediction and evolution
  - Security and self-self-awareness
- There are interesting proposals around (e.g. Knowledge Networks) but
  - A sound foundational approach is still missing
  - Definitely, plenty of room for fascinating, foundational, visionary, research!

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